The Blocking and Bootstrap Methods for Error Estimation in Monte Carlo Simulations

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Motivation

Monte Carlo simulations are central in many areas of physics

- Quantum Monte Carlo
- Lattice gauge theory
- Molecular dynamics
- Statistical mechanics

Typical goal: estimate expectation values

$$\mu = \mathbb{E}[X]$$

from a sequence of measurements

$$X_1, X_2, \ldots, X_N$$

Problem:

- Successive samples are often **correlated**
- Naive statistical error estimates become incorrect

The blocking method provides a practical solution.

Naive Error Estimate

For independent samples the sample mean is

$$\bar{X} = \frac{1}{N} \sum_{i=1}^{N} X_i$$

The variance of the estimator is

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{N}$$

with

$$\sigma^2 = \mathrm{Var}(X)$$

Naive Error Estimate, estimator

Estimator for the error:

$$\sigma^2_{\bar{X}} = \frac{1}{N(N-1)} \sum_{i=1}^{N} (X_i - \bar{X})^2$$

This formula is only correct if the samples are **independent**.

Correlated Data

In Markov Chain Monte Carlo we generate

$$X_1 \to X_2 \to X_3 \to \dots$$

via a Markov process.
Thus

$$\mathrm{Cov}(X_i, X_j) \neq 0$$

Define the autocorrelation function

$$C(t) = \langle (X_i - \mu)(X_{i+t} - \mu) \rangle$$

Correlated Data, Normalized

Normalized autocorrelation

$$\rho(t) = \frac{C(t)}{C(0)}$$

Typically

$$\rho(t) \sim e^{-t/\tau}$$

where τ is the autocorrelation time.

Variance of the Mean with Correlations

The variance of the mean becomes

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{N}\left(1 + 2\sum_{t=1}^{\infty}\rho(t)\right)$$

Define the **integrated autocorrelation time**

$$\tau_{\mathrm{int}} = \frac{1}{2} + \sum_{t=1}^{\infty}\rho(t)$$

Then

$$\mathrm{Var}(\bar{X}) = \frac{2\tau_{\mathrm{int}}\sigma^2}{N}$$

Variance of the Mean with Correlations, Interpretation

Interpretation

- Effective number of independent samples

$$N_{\mathrm{eff}} = \frac{N}{2\tau_{\mathrm{int}}}$$

Idea Behind the Blocking Method

Instead of estimating autocorrelations explicitly:

- Combine neighboring samples into blocks
- Each blocking level reduces correlations

If blocks are large enough:

$$\text{block averages} \approx \text{independent samples}$$

Then the usual variance estimator becomes valid again.

Blocking Transformation

Start with dataset

$$X_i^{(0)}, \quad i = 1, \ldots, N$$

Define the blocking transformation

$$X_i^{(k+1)} = \frac{1}{2}\left(X_{2i-1}^{(k)} + X_{2i}^{(k)}\right)$$

Blocking Transformation, Properties

Properties:

- Number of samples halves each iteration
- Correlations decrease with blocking level

After k transformations:

$$N_k = \frac{N}{2^k}$$

Variance at Blocking Level k

Compute

$$\bar{X}_k = \frac{1}{N_k} \sum_{i=1}^{N_k} X_i^{(k)}$$

Variance estimate

$$s_k^2 = \frac{1}{N_k - 1} \sum_{i=1}^{N_k} \left(X_i^{(k)} - \bar{X}_k \right)^2$$

Variance at Blocking Level k, Error estimate

Error estimate

$$\sigma_k^2 = \frac{s_k^2}{N_k}$$

As correlations disappear

$$\sigma_k^2 \to \mathrm{Var}(\bar{X})$$

Convergence of the Blocking Method

Key observation:

- Initially the variance estimate increases
- Eventually it reaches a plateau

Plateau value gives the correct error estimate.
Graphically:

- Plot variance vs blocking level
- Choose region where curve stabilizes

This plateau indicates that blocks are effectively independent.

Rigorous Derivation from the Covariance Matrix

Let

$$\boldsymbol{X} = (X_1, X_2, \ldots, X_N)^T$$

be a random vector with mean

$$\mathbb{E}[\boldsymbol{X}] = \mu \mathbf{1}, \qquad \mathbf{1} = (1, 1, \ldots, 1)^T.$$

The sample mean can be written as

$$\bar{X} = \frac{1}{N} \mathbf{1}^T \boldsymbol{X}.$$

We want the variance of $\bar{X}$ when the entries of $\boldsymbol{X}$ are correlated. Define the covariance matrix

$$\Sigma_{ij} = \mathrm{Cov}(X_i, X_j) = \mathbb{E}\big[(X_i - \mu)(X_j - \mu)\big].$$

Then

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{N} \mathbf{1}^T \boldsymbol{X}\right).$$

Variance of a Linear Combination

For any random vector $\boldsymbol{X}$ and deterministic vector $\boldsymbol{a}$,

$$\mathrm{Var}(\boldsymbol{a}^T \boldsymbol{X}) = \boldsymbol{a}^T \Sigma \boldsymbol{a}.$$

Choosing

$$\boldsymbol{a} = \frac{1}{N}\mathbf{1},$$

we obtain

$$\mathrm{Var}(\bar{X}) = \left(\frac{1}{N}\mathbf{1}\right)^T \Sigma \left(\frac{1}{N}\mathbf{1}\right) = \frac{1}{N^2}\mathbf{1}^T \Sigma \mathbf{1}.$$

Variance of a Linear Combination

Writing this out explicitly,

$$\mathrm{Var}(\bar{X}) = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \Sigma_{ij} = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \mathrm{Cov}(X_i, X_j).$$

This is the exact formula for the variance of the sample mean for correlated data.

Stationary Time Series and Toeplitz Structure

Assume the process is weakly stationary:

- $\mathbb{E}[X_i] = \mu$ independent of i
- $\mathrm{Cov}(X_i, X_j)$ depends only on the lag $|i - j|$

Stationary Time Series and Toeplitz Structure

Then

$$\Sigma_{ij} = C(|i-j|),$$

where

$$C(t) = \mathrm{Cov}(X_k, X_{k+t}).$$

Thus the covariance matrix has Toeplitz form:

$$\Sigma = \begin{pmatrix} C(0) & C(1) & C(2) & \cdots \\ C(1) & C(0) & C(1) & \cdots \\ C(2) & C(1) & C(0) & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

The diagonal entries are the ordinary variance:

$$C(0) = \sigma^2.$$

Summing the Covariances

Using stationarity,

$$\mathrm{Var}(\bar{X}) = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} C(|i-j|).$$

We now reorganize the double sum according to the lag

$$t = |i-j|.$$

For lag $t = 0$, there are N terms.
For lag $t \geq 1$, there are

$$2(N-t)$$

terms, since both (i, j) and (j, i) contribute.

Summing the Covariances

Therefore

$$\mathrm{Var}(\bar{X}) = \frac{1}{N^2}\left[NC(0) + 2\sum_{t=1}^{N-1}(N-t)C(t)\right].$$

Equivalently,

$$\mathrm{Var}(\bar{X}) = \frac{C(0)}{N} + \frac{2}{N^2}\sum_{t=1}^{N-1}(N-t)C(t).$$

Normalized Autocorrelation Function

Introduce the normalized autocorrelation function

$$\rho(t) = \frac{C(t)}{C(0)} = \frac{C(t)}{\sigma^2}.$$

Then

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{N}\left[1 + \frac{2}{N}\sum_{t=1}^{N-1}(N-t)\rho(t)\right].$$

Rewriting,

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{N}\left[1 + 2\sum_{t=1}^{N-1}\left(1 - \frac{t}{N}\right)\rho(t)\right].$$

This formula is exact for a stationary correlated sequence of finite length N.

Large-N Limit and Integrated Autocorrelation Time

Suppose the autocorrelation decays sufficiently rapidly so that

$$\sum_{t=1}^{\infty} |\rho(t)| < \infty.$$

Then for large N,

$$1 - \frac{t}{N} \approx 1 \qquad \text{for relevant lags } t,$$

and hence

$$\mathrm{Var}(\bar{X}) \approx \frac{\sigma^2}{N} \left[1 + 2 \sum_{t=1}^{\infty} \rho(t) \right].$$

Large-N Limit and Integrated Autocorrelation Time

Define the integrated autocorrelation time

$$\tau_{\text{int}} = \frac{1}{2} + \sum_{t=1}^{\infty} \rho(t).$$

Then

$$\text{Var}(\bar{X}) \approx \frac{2\tau_{\text{int}}\sigma^2}{N}.$$

This shows precisely how correlations inflate the uncertainty of the mean.

Interpretation in Terms of Effective Sample Size

If the data were independent, we would have

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{N}.$$

For correlated data,

$$\mathrm{Var}(\bar{X}) \approx \frac{2\tau_{\mathrm{int}}\sigma^2}{N}.$$

Hence we define an effective number of independent samples by

$$\frac{\sigma^2}{N_{\mathrm{eff}}} = \frac{2\tau_{\mathrm{int}}\sigma^2}{N}.$$

Therefore

$$N_{\mathrm{eff}} = \frac{N}{2\tau_{\mathrm{int}}}.$$

So autocorrelations reduce the information content of the sample from N to N_{eff}.

Why Blocking Works

The blocking transformation is

$$X_i^{(k+1)} = \frac{1}{2}\left(X_{2i-1}^{(k)} + X_{2i}^{(k)}\right).$$

This is a linear coarse-graining of the original correlated sequence. Its effect is to suppress short-range correlations and to reduce the off-diagonal structure of the covariance matrix.
At sufficiently high blocking level, the transformed covariance matrix becomes approximately diagonal:

$$\Sigma^{(k)} \approx s_k^2 I.$$

Then

$$\mathrm{Var}(\bar{X}) \approx \frac{s_k^2}{N_k},$$

which is exactly the independent-sample formula.
Thus blocking is a practical numerical procedure for converting the full covariance structure into a reliable variance estimate.

Mathematical Summary

Starting from the covariance matrix,

$$\mathrm{Var}(\bar{X}) = \frac{1}{N^2}\mathbf{1}^T\Sigma\mathbf{1}.$$

For a stationary process,

$$\Sigma_{ij} = C(|i-j|),$$

so

$$\mathrm{Var}(\bar{X}) = \frac{1}{N^2}\left[NC(0) + 2\sum_{t=1}^{N-1}(N-t)C(t)\right].$$

Mathematical Summary

In terms of $\rho(t) = C(t)/C(0)$,

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{N}\left[1 + 2\sum_{t=1}^{N-1}\left(1 - \frac{t}{N}\right)\rho(t)\right].$$

For large N,

$$\mathrm{Var}(\bar{X}) \approx \frac{2\tau_{\mathrm{int}}\sigma^2}{N}.$$

This is the theoretical foundation behind blocking analysis.

Blocking Algorithm

Given data $X_1, \ldots, X_N$:

1. Compute variance estimate

$$\sigma_0^2$$

2. Perform blocking transformation

$$X_i \to \frac{X_{2i-1} + X_{2i}}{2}$$

3. Recompute variance

$$\sigma_k^2$$

4. Repeat until

$$N_k < 2$$

5. Identify plateau region in variance

The plateau gives the best estimate of the statistical error.

Blocking Summary

- Monte Carlo samples are typically correlated
- Naive variance estimates underestimate errors
- Blocking reduces correlations through recursive averaging
- Variance converges to a plateau corresponding to correct error
- Simple, robust and widely used in computational physics

Blocking remains one of the most reliable error estimation techniques in large-scale Monte Carlo simulations.

Basic Idea of the Bootstrap

Suppose the unknown distribution generating the data is

$$F(x)$$

but we only observe samples

$$X_1, \ldots, X_N \sim F$$

The bootstrap replaces the unknown distribution by the empirical distribution

$$\hat{F}_N(x)$$

defined by

$$\hat{F}_N(x) = \frac{1}{N} \sum_{i=1}^{N} \delta(x - X_i)$$

Basic Idea of the Bootstrap, Interpretation

Interpretation:

- Treat the observed dataset as the population
- Generate new samples by resampling with replacement

This produces synthetic datasets from $\hat{F}_N$.

Bootstrap Resampling

Generate a bootstrap sample

$$X_1^*, X_2^*, \ldots, X_N^*$$

by drawing N values from the original data

$$\{X_1, \ldots, X_N\}$$

with replacement.
Example:

$$\{1.2, 0.8, 1.4, 0.9\}$$

Possible bootstrap sample:

$$\{0.8, 1.4, 1.4, 1.2\}$$

Each bootstrap dataset mimics an independent experiment.

Bootstrap Distribution of an Estimator

For each bootstrap sample compute

$$\hat{\theta}^* = f(X_1^*, \ldots, X_N^*)$$

Repeat this procedure B times:

$$\hat{\theta}_1^*, \hat{\theta}_2^*, \ldots, \hat{\theta}_B^*$$

These values form the **bootstrap distribution**

$$P_B(\hat{\theta})$$

which approximates the sampling distribution of the estimator

$$P(\hat{\theta})$$

in the limit of large N.

Bootstrap Estimate of the Variance

The bootstrap estimate of the variance is

$$\mathrm{Var}_{\text{boot}}(\hat{\theta}) = \frac{1}{B-1}\sum_{b=1}^{B}(\hat{\theta}_b^* - \bar{\theta}^*)^2$$

where

$$\bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}_b^*$$

Thus the bootstrap standard error becomes

$$\sigma_{\text{boot}} = \sqrt{\mathrm{Var}_{\text{boot}}}.$$

As $B \to \infty$, the estimate converges to the true variance.

Consistency of the Bootstrap

Let

$$\hat{\theta} = f(X_1, \ldots, X_N)$$

be an estimator.
Under mild regularity conditions,

$$\sqrt{N}(\hat{\theta} - \theta)$$

has a limiting distribution.
Bootstrap theory shows

$$\sqrt{N}(\hat{\theta}^* - \hat{\theta})$$

approximates the same limiting distribution.
Thus

$$P^*(\hat{\theta}^* - \hat{\theta}) \approx P(\hat{\theta} - \theta).$$

This is the theoretical justification of the bootstrap.

Bootstrap Confidence Intervals

Let

$$\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$$

be sorted.

Percentile interval

For confidence level $1 - \alpha$:

$$[\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^*]$$

Example:

95% interval

$$[\hat{\theta}_{2.5\%}^*, \hat{\theta}_{97.5\%}^*]$$

This method uses the empirical quantiles of the bootstrap distribution.

Bias Estimation

Bootstrap also estimates bias:

$$\text{Bias} = \mathbb{E}[\hat{\theta}] - \theta$$

Bootstrap estimate

$$\widehat{\text{Bias}} = \bar{\theta}^* - \hat{\theta}$$

Bias-corrected estimator:

$$\hat{\theta}_{BC} = \hat{\theta} - \widehat{\text{Bias}}$$

Bootstrap therefore provides both

- variance estimates
- bias corrections

Bootstrap Algorithm

Given data

$$X_1, \ldots, X_N$$

1. Choose number of resamples B
2. For $b = 1 \ldots B$
   - Draw bootstrap sample with replacement

     $$X_1^*, \ldots, X_N^*$$

   - Compute estimator

     $$\hat{\theta}_b^*$$

3. Compute

$$\sigma_{\text{boot}}$$

Typical choice:

$$B = 1000 - 10000$$

Applications in Physics

Bootstrap is widely used in

- Lattice QCD
- Quantum Monte Carlo
- Molecular simulations
- Experimental particle physics
- Parameter estimation in model fitting

Examples of estimators:

- correlation functions
- energy differences
- nonlinear observables
- fitted parameters

Example: Monte Carlo Observable

Suppose we estimate an observable

$$\langle O \rangle = \frac{1}{N} \sum_i O_i$$

But we want uncertainty for a nonlinear function

$$R = \frac{\langle A \rangle}{\langle B \rangle}$$

Analytic variance formulas become complicated.
Bootstrap procedure:

- resample pairs (A_i, B_i)
- recompute ratio
- obtain distribution of R

This yields robust uncertainty estimates.

Bootstrap vs Other Resampling Methods

Common resampling techniques:

- Bootstrap
- Jackknife
- Blocking

Bootstrap advantages:

- works for complex estimators
- easy to implement
- provides full distribution

Bootstrap vs Other Resampling Methods

Bootstrap limitation:

- assumes independent samples

For correlated Monte Carlo data we combine:

- blocking + bootstrap

Summary Bootstrap

Bootstrap resampling is a powerful statistical tool.

- Approximates sampling distributions numerically
- Requires minimal analytical assumptions
- Provides variance, bias, and confidence intervals
- Widely used in computational physics

In modern simulations bootstrap analysis is a standard component of data analysis pipelines.